Data Driven Control Integration Week

Group 8 (MPC)

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Friday, November 6, 2026

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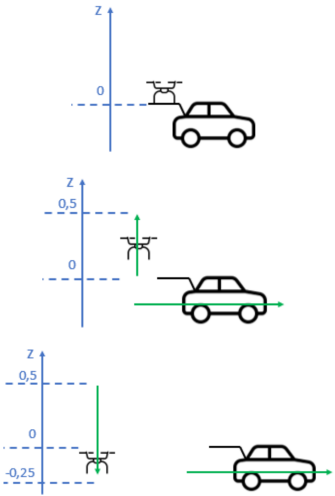
Project Objectives

Available Resources:

- Dataset of vertical acceleration.
- MATLAB/Simulink drone simulator.

Objectives:

- Validate the dataset.
- Identify the model.
- Develop a control law.



Dataset Validation

Cleaning the data

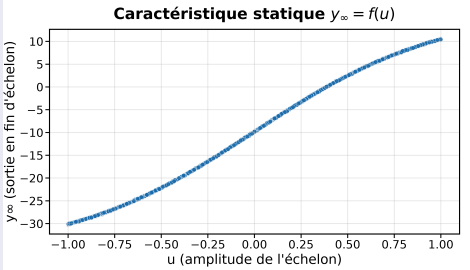
Remove lines that includes:

- Frozen sensor for 2+ seconds
- 5+ diverging values

Normalisation

- Directly in the training

Static characteristic



$y_\infty = f(u)$ after eliminating corrupted runs.

System Modeling

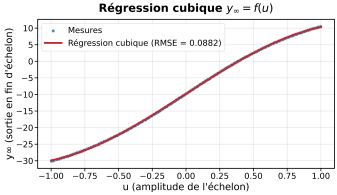
Linear Model: Hammerstein Approach

Static Characteristic $N(u)$

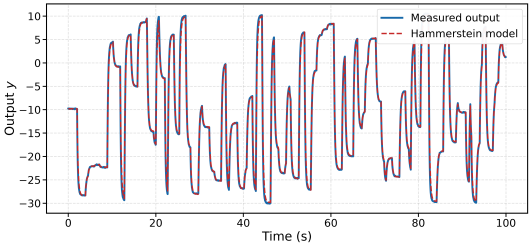
3rd-order polynomial identified on steady-state regimes:

$$v(t) = N(u) = a_3 u^3 + a_2 u^2 + a_1 u + a_0$$

Tau identification



Hammerstein Model Validation



Least-Squares Linear Model Validation (a=0.7836, b=4.9676, c=-2.1222)

System Modeling

Linear Model: Least Square Method

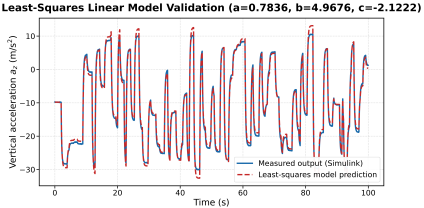
Model equations

$$y_k(t+1) = a\,y_k(t) + b\,u_k(t) + c$$
$$= \varphi_k^{\mathsf{T}}\theta.$$

$$\varphi_k^{\mathsf{T}} = \begin{bmatrix} y_k(t) & u_k(t) & 1 \end{bmatrix} \qquad \theta = \begin{bmatrix} a & b & c \end{bmatrix}^{\mathsf{T}}.$$

The least-squares estimate is

$$\theta^{\star} = \left(\Phi^{\mathsf{T}}\Phi\right)^{-1}\Phi^{\mathsf{T}}Y.$$



First RNN

Sequence-to-sequence architecture

Parameters

- One scalar input u_t and one scalar output $\hat{y}_t$
- Hidden state dimension: 16
- Early stopping to avoid overfitting

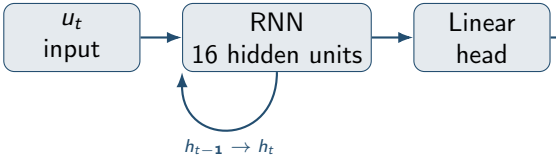
Forward equations

$$h_t = \tanh(W_{ih}u_t + W_{hh}h_{t-1} + b)$$

$$\hat{y}_t = W_{hy}h_t + b_y$$

Least-squares criterion

$$\hat{\quad} \quad \sum_{N} \quad \hat{\quad}^2$$

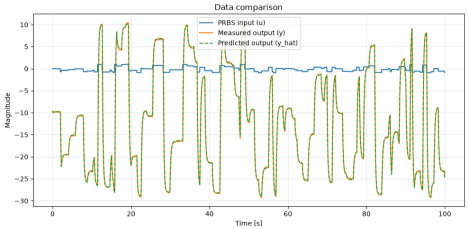


First RNN

Results

Numerical results

- Validation loss: 3.93×10^{-4}
- Test loss: 4.01×10^{-4}
- Training completed at epoch 100
- Only 321 trainable parameters



Z-axis control

Designing the command

Principle

- Python Class Wrapping
- Simulink Signal Generation
- LQ control for the linear model
- MPC control for the RNN

Results

Add here the main result or performance metric.

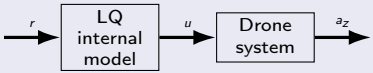
Figure

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LQ control

Linear-quadratic control design

Control principle



Performance

Add here the main performance result or metric.

